

People Analytics

Project Report

at the Faculty of Business, Economics, and Law
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1 Personnel Planning

1.1 HR Staffing Model Application (5 Points)

The **Future Number of Employees** is the expected total workforce after adjusting for retirements, resignations, and new hires. This is the actual number of employees expected to be in the organisation in the future.

The **Gross Employee Requirement (GER)** is the total number of employees the organization needs for accomplishing organizational goals and bridging skill gaps. It also includes any additional staff required to cover absenteeism, growth and so on.

Calculations

1. Train Conductors

$$\text{a. Future No. of Employees} = \text{Current Employees} - \text{Expected Attrition} + \text{Expected Hires} = 5 - 2 + 3 = \mathbf{6}$$

$$\text{b. GER} = \text{Future No. of Employees} + \text{Extra Hires} = 6 + 1 = \mathbf{7}$$

2. Bus Drivers

$$\text{a. Future No. of Employees} = \text{Current Employees} - \text{Expected Attrition} + \text{Expected Hires} = 8 - 2 + 3 = \mathbf{9}$$

$$\text{b. GER} = \text{Future No. of Employees} + \text{Extra Hires} = 9 + 2 = \mathbf{11}$$

3. Maintenance Technicians

$$\text{a. Future No. of Employees} = \text{Current Employees} - \text{Expected Attrition} + \text{Expected Hires} = 4 - 1 + 2 = \mathbf{5}$$

$$\text{b. GER} = \text{Future No. of Employees} + \text{Extra Hires} = 5 + 1 = \mathbf{6}$$

1.2 Optimal Staff Need (5 Points)

The objective was to optimize staffing requirements for conductors across four shifts based on passenger demand and wage costs. The initial step involved loading the passenger data and wage rates for each shift from the dataset. Passenger counts for each shift were aggregated by identifying which shift each data entry belonged to and collecting the average passenger numbers accordingly. Wage rates were assigned to each shift from the corresponding dataset row.

To model the problem, a linear programming approach was adopted, defining decision variables representing the number of conductors needed per shift. The objective function aimed to minimize the total wage cost, calculated as the sum of conductors multiplied by their respective shift wages.

Constraints ensured that the number of conductors assigned to each shift was sufficient to serve peak passenger demand. Specifically, the maximum number of passengers observed in any time slot within each shift was used to determine the minimum required conductors,

assuming each conductor can handle 10 passengers. The staffing variables were constrained to be at least this calculated number.

The model was solved using an LP solver to find the minimum cost staffing plan that meets service level requirements. The solution provides the optimal number of conductors to schedule per shift while minimizing overall wage expenses.

Optimisation Problem

Decision Variables

x_1 = Optimal staff needed for Shift 1
 x_2 = Optimal staff needed for Shift 2
 x_3 = Optimal staff needed for Shift 3
 x_4 = Optimal staff needed for Shift 4

```
[165]: # Creating the variable for the optimisation problem
shifts_decision_vars = LpVariable.dicts("Fau_Transline", shift_details.keys(), lowBound=0, cat='Integer')
```

Objective Function

Minimize: $60x_1 + 45x_2 + 45x_3 + 70x_4$

```
[166]: # Defining the Problem
prob = LpProblem("Optimal_Staff_Need", LpMinimize)

# Defining the Objective Function
prob += lpSum(shift_details[shift]['wage'] * shifts_decision_vars[shift] for shift in shift_details), "Total_Wage_Cost"
```

Constraints

$$\text{Service Level} = \frac{1}{10} = 0.1$$

$$x_1 \geq \lceil \text{max(average number of people per hour in shift 1)} \times 0.1 \rceil$$

$$x_2 \geq \lceil \text{max(average number of people per hour in shift 2)} \times 0.1 \rceil$$

$$x_3 \geq \lceil \text{max(average number of people per hour in shift 3)} \times 0.1 \rceil$$

$$x_4 \geq \lceil \text{max(average number of people per hour in shift 4)} \times 0.1 \rceil$$

```
[167]: # Defining the Constraints
service_level = 1/10
for shift, details in shift_details.items():
    max_passengers = max(details['passengers'])
    required_staff = math.ceil(max_passengers * service_level)
    prob += shifts_decision_vars[shift] >= required_staff, f"{shift}_Constraint"
```

▼ Solving the Problem and Displaying the Results

```
[170]: prob.solve()
print("Status:", LpStatus[prob.status])
for variable in prob.variables():
    print("The number of workers needed for", variable.name, " are ", int(variable.varValue))
```

```
Status: Optimal
The number of workers needed for Fau_Transline_Shift_1 are 10
The number of workers needed for Fau_Transline_Shift_2 are 5
The number of workers needed for Fau_Transline_Shift_3 are 5
The number of workers needed for Fau_Transline_Shift_4 are 1
```

1.3 Reskilling vs. External Hiring (4 Points)

Trade-offs Between Reskilling and External Hiring in the context of Public Transportation.

Reskilling

Reskilling means helping current employees learn new skills so they can move into different roles within the company. This can save money, since hiring new staff externally usually comes with higher salary demands. Reskilled employees also understand the company's systems and culture, which means they need less training to get up to speed. Providing growth opportunities through training can also boost motivation and help retain valuable staff.

However, reskilling is not always practical. Some jobs like bus conductors require people skills, safety training, or licensing that employees in other departments (like maintenance technicians or drivers) might not have or even want to pursue. In fact, someone trained in technical repairs may not be interested in working directly with passengers or handling service-related duties. Also, reskilling takes time, which might not be ideal when staff shortages need to be addressed quickly, such as during peak travel periods or when older employees retire.

External Hiring

External recruitment allows the organization to bring in candidates who already have the necessary skills and qualifications for specific roles. This is especially useful when there is an urgent need to fill positions such as during service expansion or staff retirements. And, when internal candidates are not readily available. External hires can also bring fresh ideas, diverse experiences, and best practices from other organizations, which can drive innovation and improvement over time.

However, hiring from outside is often more expensive. These candidates typically expect higher salaries, and additional resources are needed for onboarding and training. It can also take time for new employees to adapt to the organization's culture and processes. In public transportation, where consistency and reliability are essential, there's also a higher risk that new hires may leave if expectations or working conditions don't align with their experience.

Recommended Approach for FAU Transline

Given FAU Transline's current staffing gaps particularly the shortage of conductors needed to maintain service levels and the urgency created by retirements and rising passenger demand, relying solely on **external hiring appears to be the most practical solution**. Reskilling would require time, training infrastructure, and interest from current employees, which may not align with immediate operational needs. While reskilling offers long-term benefits such as cost savings and improved employee morale, its effectiveness depends heavily on role compatibility. For instance, bus drivers may be reskilled into train conductor roles, given their

similar operational and customer-facing experience. However, maintenance technicians may struggle to adapt, as the interpersonal and regulatory skills required differ significantly from their technical background. Therefore, external recruitment remains the fastest and most effective path forward to ensure continuity and maintain service quality.

2 Sourcing and Acquisition

2.1 Personnel selection process (4 Points)

Before hiring new employees, FAU Transline first needs to understand what makes a good worker for each role. This process is called requirements analysis, and it's an important step to make sure the right people are hired later on.

One method is the experiential-intuitive approach. This involves talking to people who already do these jobs or manage them. These experienced insiders really know the day-to-day realities and what skills are actually needed to do the job well. Their practical knowledge helps create job descriptions that match what the role truly involves, not just what's written on paper. This experience-based insight is a solid starting point because it keeps things grounded in reality.

Another important method is the empirical approach, which has two parts. The “workplace analytical empirical” approach uses standardized tools like questionnaires or observations to analyze job tasks and requirements in detail. For example, by using questionnaires or watching how tasks are done, FAU Transline can break down exactly what each job requires. Meanwhile, the “personal-empirical” approach looks at statistics about employee traits and how they relate to success in a role helps identify which qualities make someone a good fit. Combining these data-driven methods means the company can hire people who match the job's real demands, rather than relying on guesswork.

This kind of job analysis is essential because it sets the stage for everything else in recruitment. By blending real-world experience with solid data, FAU Transline can build clear and accurate profiles for each role. This makes hiring smarter and helps the company find employees who will succeed and stick around.

2.2 Data driven recruitment (7 Points)

FAU Transline is thinking about using data-driven tools to help with its recruitment process. There are several clear benefits to this approach when it comes to selecting applicants. Hiring can be expensive, slow, and risky, especially if the wrong person is chosen. By automating parts of recruitment, FAU Transline can save money, speed up hiring, and make better decisions overall.

Data-driven recruitment tools work by analyzing large amounts of applicant information such as resumes, online behavior, and skills. This allows the system to quickly identify candidates who are the best fit for the job. Automating tasks like resume screening and interview scheduling reduces the workload for HR and shortens the hiring timeline.

More importantly, these tools help ensure that decisions are based on objective data related to the job rather than personal feelings or biases. This leads to better matches between

candidates and roles and improves employee retention. According to Marks Sattin (2025), data-driven hiring can cut costs and improve the quality of hires significantly.

For a public transportation company like FAU Transline, which may need to grow its workforce quickly and responsibly, data-driven recruitment tools offer real advantages.

Steps to identify the relevant association rules

To identify association rules predicting successful hiring for the *intercity bus driver* role

1. Data Preprocessing
 - a. Dropped columns like depot_driver and city_bus_driver because they were unrelated target roles. Also removed recruitment_strategy and distance_from_company as they do not directly reflect candidate skills or qualifications.
 - b. Retained key input features focusing on skills, education level, experience, and previous companies, which are more relevant for predicting hiring outcomes.
2. Feature Binning
 - a. Binned numeric features into categories to improve interpretability:
 - i. experience_years: Novice [0-3], Skilled [3-7], Advanced [7-10], Expert [10-15]
 - ii. safe_driving_skills, customer_service_skills, navigation_skills: Poor [0-60], Avg [60-75], Good [75-85], Best [85-100]
 - iii. previous_companies: [0-1], [2-3], [4-5]
3. One-Hot Encoding
 - a. Transformed all categorical/binning features into binary columns using `pd.get_dummies`.
4. Applying the Apriori Algorithm
 - a. Used the `apriori()` function with `min_support = 0.01` to identify frequent itemsets and to get an optimal number of rules.
 - b. Then used `association_rules()` with `min_confidence = 0.5` to generate relevant rules.
5. Filtering and Sorting
 - a. Filtered rules where the consequent (outcome) was {'intercity_bus_driver', 'hiring_decision'}.
 - b. Removed rules where the antecedents contained outcome labels 'intercity_bus_driver', 'hiring_decision'.
 - c. Sorted the rules by lift to prioritize the most significant associations.

The key skills, qualifications, and patterns strongly linked to being hired for the intercity bus driver role

Based on the top 10 association rules (sorted by lift), successful hiring is strongly linked to specific combinations of education level, experience years, and skills:

- The strongest pattern is candidates with education level realschule or higher, good safe driving skills (75–85), and good customer service skills (75–85).
- Another common combination is education level realschule or higher, advanced experience (7–10 years), and either good safe driving skills (75–85) or good customer service skills (75–85).
- Expert experience (10–15 years) combined with realschule or higher education and good customer service skills (75–85) also appears as a strong pattern.
- Candidates with hauptschule education combined with good safe driving skills (75–85) or good navigation skills (75–85) form another relevant group.
- Combinations involving navigation skills from average (60–75) to best (85–100) alongside good safe driving skills (75–85) and good customer service skills (75–85) also appear in the patterns.

Metrics for the Top Rule of Association Analysis

The top rule highlights candidates possessing a combination of education at the realschule level or higher, along with good safe driving skills (rated 75–85) and good customer service skills (also rated 75–85). The **support** value of 0.010 means that this specific profile appears in 1% of the entire candidate dataset. With a **confidence** of 0.938, it indicates that 93.8% of candidates matching this profile were hired, reflecting a very strong likelihood of selection based on these traits. Moreover, the **lift** value of 3.111 suggests that candidates with this combination are over three times more likely to be hired than if hiring decisions were made randomly, underscoring the significant influence of this skill and education mix on hiring outcomes.

2.3 Ethical Risks in automation (4 Points)

If FAU Transline introduces an AI-based hiring tool, it should be aware of several ethical and legal risks. These include potential bias and discrimination, lack of transparency in decision-making, privacy concerns related to handling sensitive applicant data, and the possibility of violating labor and anti-discrimination laws. Such risks can undermine fairness, damage the company's reputation, and lead to legal consequences.

For instance, Amazon scrapped its AI recruiting tool after discovering it discriminated against female applicants due to biased training data (BBC, 2018).

Another study highlighted significant racial, gender, and intersectional biases in how large language models ranked resumes, favoring white-associated names 85% of the time, female-associated names only 11% of the time, and never favoring Black male-associated names over white male-associated names.

To reduce potential bias in automated hiring, FAU Transline should ensure their AI models are trained on diverse and representative datasets to prevent reinforcing existing biases. They should regularly audit the hiring outcomes to detect and correct discriminatory patterns. Transparency in the AI's decision-making process is crucial so that human recruiters can understand and challenge automated recommendations. Additionally, combining AI tools with human judgment helps maintain fairness by considering context beyond what algorithms capture. Finally, focusing AI assessments strictly on job-relevant criteria minimizes the influence of irrelevant or biased factors (Harvard Business Review, 2019).

3 Onboarding and Performance

3.1 A new hire (5 Points)

Socialization is essential in onboarding as it helps new employees understand the organization's mission, values, and culture, aligning their personal goals with those of the company. This alignment fosters person–organization fit, which enhances early role adjustment and long-term success. Effective socialization leads to increased productivity, job satisfaction, organizational commitment, and longer tenure (Feldman, 1981; Fisher, 1986).

According to Bauer (2010), effective onboarding includes four key components, known as the 4 C's:

Connection

This involves helping new employees build relationships and information networks. FAU Transline can support this by introducing buddy systems, encouraging team meetups, and using social recommender systems to connect new hires with colleagues.

Clarification

New employees need to understand their job roles and expectations. FAU Transline can provide this through regular conversations with supervisors, initial interviews, and internal tools like enterprise social networks to ensure ongoing clarity.

Culture

Employees should learn about the organization's values, norms, and ways of working. FAU Transline can share this through knowledge wikis and internal platforms that explain both formal procedures and informal workplace culture.

Compliance

This covers legal rules, company policies, and safety standards. FAU Transline can deliver this information through onboarding workshops and digital resources like policy wikis for easy access.

FAU Transline should use the following socialization tactics based on Van Maanen and Schein's framework:

- **Collective:** Organize group onboarding sessions so new employees can learn together and start building relationships.
- **Formal:** Provide structured onboarding events separate from daily work, like dedicated training sessions.
- **Sequential:** Plan a clear, step-by-step onboarding process, such as safety training first, followed by role-specific learning and then supervised practice.

- **Fixed:** Set a defined schedule for onboarding activities so new hires know what to expect and when.
- **Serial:** Assign experienced mentors to guide new employees through the onboarding process.
- **Investiture:** Focus on giving positive feedback to build confidence and support new employees' strengths.

Successful onboarding helps build strong working relationships, which improve communication and cooperation among colleagues. This leads to better performance and greater team integration. Social connections also give new employees support and a sense of belonging, reducing anxiety during the adjustment period.

Onboarding increases organizational commitment, meaning employees feel more attached to the company's values, want to stay longer, and are motivated to contribute. Employees who experience structured onboarding with clear expectations and positive feedback tend to have higher job satisfaction and feel more confident in their roles.

Research shows that employees who clearly understand their roles and feel accepted by coworkers perform better. Early socialization builds connections that serve as valuable social capital to support job performance. Studies also find that onboarding approaches involving sequential steps, fixed timelines, mentorship, and positive reinforcement are linked to stronger commitment, better person-organization fit, and higher engagement.

In short, effective onboarding improves employee performance, satisfaction, and long-term loyalty to the organization.

3.2 Recommender System (4 Points)

Profile of klara_007

id	teams	previous_experience	hobbies	sports	personality_traits
klara_007	team_03	Competent	listening to podcasts journaling gardening	table tennis	creative innovator

Top 3 Recommended Colleagues

id	teams	previous_experience	hobbies	sports	personality_traits
emma_042	team_03	Advanced beginner	walking journaling listening to podcasts	table tennis	empathetic innovator
christian_049	team_03	Competent	listening to podcasts journaling puzzle-solving	table tennis	cooperative empathetic
simon_001	team_02	Expert	reading maps walking listening to podcasts	table tennis	curious energetic

The top three recommended colleagues were chosen because they have a lot in common with klara_007. Emma_042 and Christian_049 are on the same team as klara_007, which means they are more likely to work closely together. All three enjoy listening to podcasts and journaling, so they share similar interests. They also play table tennis, which can help build social connections outside of work tasks. Their personality traits, such as being innovative, empathetic, and curious, align well with klara_007's creative and innovative nature. These

similarities suggest that klara_007 will likely find it easier to connect and collaborate with them, helping her settle in faster.

3.3 Factors that affect employee performance (4 Points)

Research shows that employee performance depends mainly on three groups of factors:

1. Firm and Environment Factors:

This includes management support, training culture, and the overall work environment. At FAU Transline, strong support from supervisors and a positive organizational climate help employees feel motivated and committed.

2. Job-Related Factors:

These are aspects like job autonomy, communication, and the work environment itself. When FAU Transline employees have freedom in their tasks and clear communication, they tend to perform better.

3. Employee-Related Factors:

These include motivation, adaptability, proactivity, skill flexibility, and commitment. Employees at FAU Transline who are motivated and adaptable can handle challenges more effectively, leading to higher performance.

Leadership style also plays a key role. Transformational leaders who inspire, empower, and support their team boost employee engagement and performance.

The technician's poor performance and early departure can be traced to several interrelated factors linked to leadership and the work environment. Research shows that management support and organizational climate strongly influence employee motivation, commitment, and proactivity. An unsupportive or unclear work climate, along with limited communication and job autonomy, can reduce engagement. Effective leadership plays a key role in shaping these factors by fostering a positive environment, encouraging open communication, and empowering employees. Thus, the supervisor's inadequate leadership likely contributed to the technician feeling unmotivated and unsupported, leading to their decision to quit after only three months.

Without effective leadership at FAU Transline, employees like the new technician feel unmotivated and may leave prematurely, causing high turnover. This disrupts team cohesion and workflow, lowers overall productivity, and increases costs for hiring and training new staff. Over time, these issues harm the organization's stability and its ability to meet operational goals efficiently.

FAU Transline can strengthen employee performance during the first year by ensuring supervisors provide strong, transformational leadership that motivates and empowers employees. Management support must be visible to build commitment and proactivity, while clear job communication and autonomy foster motivation and efficiency. Creating a positive

organizational climate with a culture of continuous training enhances employees' skills and adaptability. Encouraging teamwork and social support further improves engagement. Together, these actions reduce turnover and improve overall performance.

3.4 Employee performance analysis (8 Points)

FAU Transline's management has expressed concerns about employee performance. This project aims to analyze the current employee dataset to uncover the core causes of performance issues and develop a machine learning model to predict employee performance ratings accurately. The analysis covers exploratory data analysis (EDA), data preprocessing, correlation analysis, model selection, training, and evaluation.

Exploratory Data Analysis (EDA)

The initial step in analyzing the dataset involved checking for any missing or null values. Fortunately, the dataset was found to be complete, with no missing entries, eliminating the need for any data imputation or removal of records. This completeness ensures the reliability of subsequent analyses without concerns about data loss or bias introduced by filling gaps.

Next, irrelevant features were identified and handled appropriately. The **EmpNumber** column, which serves as a unique identifier for each employee, was dropped from the dataset. Since it does not provide any meaningful information about employee performance and is unique to every individual, retaining it could lead the model to overfit by memorizing these identifiers rather than learning generalizable patterns. All other features related to employee demographics, job roles, and work conditions were preserved, as they contain valuable information pertinent to performance analysis.

Feature Engineering and Data Transformation

To prepare the data for machine learning algorithms, categorical variables required conversion into numerical formats. This is a critical preprocessing step because most models expect numerical inputs and cannot handle text categories directly.

Binary categorical variables, such as **Gender**, **OverTimeHours**, and **Attrition**, were label encoded. This means their two categories were mapped to 0 and 1, preserving their binary nature without implying any order or ranking between the categories.

For features with multiple categories—such as **EmpJobRole** and **MaritalStatus**—one-hot encoding was applied. This technique generates separate binary indicator columns for each category within a variable, ensuring the model treats each category independently without assuming any ordinal relationship. To prevent multicollinearity among these new features, the first category of each set was dropped (`drop_first=True`), a standard practice to avoid redundant information.

These transformations effectively prepare the dataset for model training by enabling algorithms to interpret categorical data numerically while preserving the inherent meaning of each variable.

Correlation Analysis with Performance Rating

A correlation matrix was computed and visualized using a heatmap to understand the relationships among variables and their associations with the target variable, **PerformanceRating**.

The analysis revealed that **Employee Environment Satisfaction** has the strongest positive correlation with performance, suggesting that a better work environment significantly enhances employee output. Other positively correlated factors include **Salary Hike Percent**, **Job Involvement**, **Job Satisfaction**, and **Work-Life Balance**. These findings highlight that employee engagement and rewards are key drivers of higher performance.

Conversely, variables such as **Years Since Last Promotion** and **Attrition** showed negative correlations with performance. This indicates that employees who have waited longer for promotions or those who are likely to leave the company tend to perform worse.

Interestingly, demographic variables like age, gender, and job role exhibited minimal or no clear correlation with performance, suggesting that performance is less influenced by these attributes in the current dataset.

Overall, this analysis underscores the importance of workplace satisfaction and timely recognition in fostering high employee performance

Problem Definition: Classification Over Regression

While **PerformanceRating** is numerically encoded with values ranging from 1 to 4, these numbers represent discrete, ordinal categories rather than continuous quantities. Therefore, treating this as a regression problem to predict continuous values would be inappropriate and potentially misleading.

Instead, the problem is more suitably framed as a **classification task**, where the goal is to predict one of the discrete performance classes. This approach aligns with real-world needs to categorize employees into performance levels and enables the use of evaluation metrics like accuracy, precision, recall, and F1-score that are meaningful for classification.

Model Selection

Several supervised classification algorithms were considered for predicting employee performance, including Support Vector Machines, Logistic Regression, Gradient Boosting (XGBoost), K-Nearest Neighbors, Neural Networks, and Random Forest.

Given the dataset size (632 records) and the need for a model that balances predictive performance, robustness, and interpretability, the **Random Forest** classifier was selected. This ensemble method effectively handles complex, nonlinear relationships and mixed data types without extensive hyperparameter tuning. Moreover, Random Forest models naturally provide feature importance insights, aiding interpretability and supporting actionable business decisions.

Model Performance and Evaluation

The Random Forest model achieved an overall accuracy of **92%** in predicting employee performance ratings, demonstrating strong predictive capability.

Accuracy: 0.92

Classification Report

	precision	recall	f1-score	support
2	1.00	0.76	0.86	29
3	0.89	1.00	0.94	84
4	1.00	0.79	0.88	14
accuracy			0.92	127
macro avg	0.96	0.85	0.90	127
weighted avg	0.93	0.92	0.92	127

Class-wise evaluation showed:

- **Class 3** (the largest group) was predicted with high precision (89%) and perfect recall (100%), indicating the model correctly identifies nearly all employees belonging to this category.
- **Class 2** achieved perfect precision (100%) but lower recall (76%), meaning predictions for this class are highly accurate but some true instances are missed.
- **Class 4** also had perfect precision (100%) with recall at 79%, indicating reliable predictions but some cases were not detected.

The weighted F1-score of 0.92 reflects a good balance between precision and recall across all classes.

Overall, the model is reliable, especially for the majority class, but might benefit from more data to better predict less frequent performance ratings.

This project successfully analyzed employee data to uncover key factors influencing performance and developed a predictive model with high accuracy. The findings highlight that employee satisfaction with their work environment, recognition through salary hikes, and job involvement are critical to driving better performance. The Random Forest classifier provides a reliable tool for FAU Transline to anticipate employee performance ratings, supporting data-driven decisions to enhance workforce productivity.

References

- Laumer, S. (n.d.). *People analytics: Data science for human resources management* [Course materials]. Friedrich-Alexander University Erlangen-Nuremberg, Institute for Information Systems.
- PuLP Documentation Team. (n.d.). *A blending problem*. PuLP 2.8 documentation. COIN-OR. https://coin-or.github.io/pulp/CaseStudies/a_blending_problem.html
- Wikipedia contributors. (2024, May 15). *Job analysis*. Wikipedia. https://en.wikipedia.org/wiki/Job_analysis
- Hall, E. (2025, February 13). *How to use data-driven recruitment to make better hiring decisions*. Marks Sattin. <https://www.markssattin.co.uk/general/2025-2/how-to-use-data-driven-recruitment>
- GeeksforGeeks. (n.d.). *Implementing Apriori algorithm in Python*. GeeksforGeeks. <https://www.geeksforgeeks.org/machine-learning/implementing-apriori-algorithm-in-python/>
- DataCamp. (n.d.). *Apriori algorithm: How does it work?* DataCamp. <https://www.datacamp.com/tutorial/apriori-algorithm>
- Pandas Development Team. (n.d.). *pandas.get_dummies* [Documentation]. pandas. https://pandas.pydata.org/docs/reference/api/pandas.get_dummies.html
- Pandas Development Team. (n.d.). *pandas.cut* [Documentation]. pandas. <https://pandas.pydata.org/docs/reference/api/pandas.cut.html>
- BBC News. (2018). *Amazon scraps secret AI recruiting tool that showed bias against women*. <https://www.bbc.com/news/technology-45809919>
- Milne, S. (2024, October 31). *AI tools show biases in ranking job applicants' names according to perceived race and gender*. University of Washington. <https://www.washington.edu/news/2024/10/31/ai-bias-resume-screening-race-gender/>
- Harvard Business Review. (2020, January 14). *How to Reduce Bias in AI Hiring*. <https://hbr.org/2019/10/using-ai-to-eliminate-bias-from-hiring>
- Jain, A. (2020, March 23). *TF-IDF vectorization with cosine similarity*. Medium. <https://medium.com/@anurag-jain/tf-idf-vectorization-with-cosine-similarity-eca3386d4423>
- scikit-learn developers. (2023). *scikit-learn API reference*. scikit-learn. <https://scikit-learn.org/stable/api/index.html>